

Week 10 — Lab Handout (Student)

Manual QA & Bug Hunting

Time: ~90 min in-class **Work as:** Your team (5 members) **AI use this week:** ✓ On for test scaffolding — log prompts. But *you* design the QA. **This week:** Feature freeze — find and log bugs, don't fix them yet.

Why this lab matters

Today you stop building and start breaking. A systematic QA pass now means no nasty surprises during the final defense. You'll test every feature against every scenario, try hard to break the app, and produce a clean, triaged bug list that next week's fixes will work through.

Task 1 — Build your test matrix (20 min)

Lay out features down, scenarios across (happy / boundary / invalid / empty / permissions).

- ☐ Every feature has a row; every relevant scenario a column.
- ☐ Save it to `/docs/test-matrix.md`.

Task 2 — Run manual QA passes (20 min)

Work through the matrix cell by cell.

- ☐ Test each cell; mark pass/fail.
- ☐ For every failure, note what happened (you'll log it as a bug in Task 5).

Task 3 — Expand automated coverage (15 min)

- ☐ Add automated tests for the critical paths and validation rules the matrix shows are uncovered.
- ☐ Keep the suite green. Target the paths that would hurt if they broke — not 100%.

Task 4 — Break your app (15 min)

Adversarial session — try hard to break it:

- ☐ Weird input (huge numbers, emoji, empty, `<script>`).
- ☐ Out-of-order actions (back button, double-click, refresh mid-submit).
- ☐ Direct URL to a non-existent record.
- ☐ Act on something you just deleted.
- ☐ Network off / slow — does it degrade gracefully?

Task 5 — Log & triage bugs (10 min)

For every bug found:

- ☐ Title, numbered steps to reproduce, expected vs. actual, severity.
- ☐ Label **P0** (broken core / data loss), **P1** (significant, has workaround), or **P2** (minor).
- ☐ File each as an issue in your tracker.

End-of-Lab Checklist

- ☐ `/docs/test-matrix.md` — features × scenarios, marked pass/fail
- ☐ Automated tests expanded on critical gaps; suite green
- ☐ Adversarial testing session done
- ☐ Every bug logged with reproduction steps + P0/P1/P2 label
- ☐ Prompt logs current in `/docs/ai-notes/week-10.md`

Looking ahead to Week 11

You'll work the triaged bug list — fixing P0/P1 first — pay down a little technical debt, and **deploy the app to a live host**, so it finally leaves localhost.